Tanyim Hassan Auvy

Computer Science Student | Software Developer | RAG Researcher

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PROFESSIONAL SUMMARY

Computer Science student at BRAC University with project experience across full-stack web development, Android applications, computer graphics, systems programming, and database tools. Current thesis work focuses on reinforcement-learning-based retrieval triggering for Retrieval-Augmented Generation. Interested in building efficient AI-enabled applications for mobile and resource-constrained devices.

EDUCATION

B.Sc. in Computer Science - BRAC University

Expected January 2027

Relevant interests: data structures and algorithms, database systems, software engineering, large language models, Retrieval-Augmented Generation, and reinforcement learning.

RESEARCH EXPERIENCE

Reward-based Retrieval Triggering for Large Language Models

Ongoing thesis research | 2025-2026 | Supervised by Dr. Swakkhar Shatabda

Developing an RL-enhanced RAG framework that learns when retrieval is necessary instead of relying only on fixed rules. Planned evaluation uses 2WikiMultihopQA, HotpotQA, IIRC, and StrategyQA with LLaMA-2-Chat and Vicuna model families.

- Coding and experiment implementation
- Model training and evaluation support
- LaTeX editing and research poster design

Research draft

TECHNICAL SKILLS

Programming	C, C#, Java, Python, Kotlin, JavaScript, Bash
Web	Vue.js, React, Express.js, Flask API, HTML, CSS
Databases	PostgreSQL, MariaDB, MySQL, DBeaver, DBMS
Research & Tools	LLMs, RAG, Reinforcement Learning, DSA, Linux, Docker, Android Studio, OpenGL

SELECTED PROJECTS

MohollarDokan

Cross-platform Commerce App | Android & Desktop

Bilingual English/Bangla shopping app with owner inventory management, customer shopping flows, and multi-shop checkout. OpenStreetMap shop and delivery maps with server-side delivery-radius validation and per-owner data isolation.

Stack: Kotlin Multiplatform, Compose, Ktor, SQLite, OpenStreetMap • Repository

SmartGeo

Location-based Android App

OpenStreetMap app for GPS-tagged landmarks and visits, backed by a shared Kotlin Multiplatform/Ktor module. Room caching and WorkManager persist visit submissions and retry background polling when connectivity returns.

Stack: Kotlin, Jetpack Compose, Ktor, Room, WorkManager, OpenStreetMap • Repository

WelcomeSQL

Solo | Full Stack | Web Development

Interactive SQL learning platform for querying, filtering, joining, aggregating, creating tables, and modifying data.

Stack: Flask API, Vue.js, MySQL • Repository

EveSync

Solo | Full Stack | Android App

Menstrual-cycle tracking application with period logging, cycle prediction, and practical guidance.

Stack: Kotlin, Android • Repository

SeaSurf 3D

Team of 3 | Full Stack | 3D Game

Academic surfing simulator featuring real-time 3D graphics and interactive game mechanics.

Stack: Python, OpenGL • Repository

InternLink

Team of 4 | Backend & Team Lead | Web Platform

Internship management system connecting students, employers, and administrators through structured application workflows.

Stack: JavaScript, Vue.js, PostgreSQL • Repository

VangtiChai

Solo | Full Stack | Cross-platform Mobile App

Money-change calculator for quickly computing practical note and coin combinations.

Stack: Kotlin Multiplatform • Repository

VS File System

Team of 3 | Code & Diagrams | Systems Programming

Very-simple file-system image generator supported by low-level filesystem structures and diagrams.

Stack: C, File Systems • Repository

Tango

Solo | Full Stack | Puzzle Game

Python puzzle game designed around logical thinking, interaction, and problem solving.

Stack: Python • Repository

CAREER DIRECTION

Preferred fields: LLM research, software development, and computer science instruction.

Research interests: Retrieval-Augmented Generation, reinforcement learning, and advanced algorithms.

Future study: M.Engg. in CSE, with interest in European Union-affiliated programs. **Additional languages:** Spanish and Turkish.